

Advanced Financial Modeling Course 26/27

Mock Exam

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Questions

1. Interest Rate Derivatives

You are managing a 2-year Receiver Swap on a \$1M Notional, the fixed leg is based on a 5% annual rate, the floating leg pays 1-year Libor annually. The yield curve is perfectly flat at 5%.

- (a) Assume the yield curve undergoes an instantaneous parallel shift of +1 basis point, immediately after the swap is signed, but before the first Libor rate is fixed for the next period. Calculate the new value of the Floating Leg.
- (b) Now, calculate the impact of that same shift on the Fixed Leg.
- (c) The DV01 of the Swap is defined as:

$$DV01_{Swap} = DV01_{Fixed_Leg} - DV01_{Float_Leg}$$

Based on part (a) and (b) results, which leg "dominates" the DV01 of the swap ?

Solution:

- (a) Calculate the initial value ($y = 5\%$). In a flat 5% environment, the floating rate for both year 1 and year 2 is 5%.

$$PV_{float} = \frac{50,000}{(1.05)^1} + \frac{1,050,000}{(1.05)^2} = 47,619.05 + 952,380.95 = 1,000,000$$

Apply the +1bp shift ($y = 5.01\%$). Because the Libor rates for future periods reset to the new market rates, the coupons also increase to 5.01%.

$$PV_{float,new} = \frac{50,100}{(1.0501)^1} + \frac{1,050,100}{(1.0501)^2}$$
$$PV_{float,new} = 47,709.74 + 952,290.26 = 1,000,000$$

The DV01 of the floating leg is 0; because the numerator (coupon) and denominator (discount rate) moved in tandem, the leg stays at par.

- (b) Calculate the initial value ($y = 5\%$). The fixed leg receives 5% regardless of market moves.

$$PV_{fixed} = \frac{50,000}{(1.05)^1} + \frac{1,050,000}{(1.05)^2} = 1,000,000$$

Apply the +1bp shift ($y = 5.01\%$). The coupons remain at 5%, but the discount rate increases.

$$PV_{fixed,new} = \frac{50,000}{(1.0501)^1} + \frac{1,050,000}{(1.0501)^2}$$

$$PV_{fixed,new} = 47,614.51 + 952,199.79 = 999,814.30$$

$$\Delta PV = 1,000,000 - 999,814.30 = 185.70$$

- (c) Combining the results

$$DV01_{Swap} = DV01_{Fixed} - DV01_{Float} = 185.70 - 0 = 185.70$$

This represents the dollar value change for every 1bp parallel shift in the curve. If you are a Receiver (Long Fixed), your PV dropped when rates went up. You have a positive DV01 (long duration). If you are a Payer (Short Fixed), your PV increased (relative to the market), giving you a negative DV01. Note that in the real world, the floating leg DV01 isn't exactly zero because of the "accrued interest" since the last reset date.

2. Change of Numeraire

Consider a derivative contract with a unit notional that pays at time $T = 2$. The payoff is based on the natural logarithm of the forward rate resetting at $T = 1$, minus a strike K :

$$\text{Payoff} = (\ln(F(1; 1; 2)) - K)$$

- (a) Solve the following pricing problem, discuss the underlying hypotheses, the choice of numeraire, and the mathematical tools required. Why is the $T = 2$ forward measure the logic choice and what can be assumed on the forward rate $F(t; 1; 2)$.
- (b) Does the price of this contract depend on the volatility σ ? If so, describe the relationship and identify a market instrument from which you could calibrate this parameter.

Solution:

- (a) The choice of the Zero-Coupon Bond $P(t; 2)$ as the numeraire is the most logical since the payment occurs at $T = 2$. The forward rate $F(t) = F(t; 1; 2)$ is a martingale under the forward measure Q^{T^2} so

$$dF(t; 1; 2) = \sigma F(t; 1; 2) dW_t^{T^2}$$

We can use Itô's Lemma to find the dynamics of $\ln(F)$ and the property that under Q^{T^2} . The current forward price is the discounted expected value of the payoff

$$\text{Price}(0; L(1; 2); K) = \mathbb{E}^Q [D(0; 2) (\ln(F(1; 1; 2)) - K)]$$

From our assumptions the solution for $F(1)$ is:

$$F(1) = F(0) \exp \left(-\frac{1}{2} \sigma^2 (1) + \sigma W_1^{T^2} \right)$$

take the natural log:

$$\ln(F(1)) = \ln(F(0)) - \frac{1}{2}\sigma^2 + \sigma W_1^{T^2}$$

Hence the price at $t = 0$ is:

$$\begin{aligned} \text{Price} &= P(0; 2) \mathbb{E}^{Q^{T^2}} [\ln(F(1)) - K] \\ \text{Price} &= P(0; 2) \left(\ln(F(0)) - \frac{1}{2}\sigma^2 + \sigma \mathbb{E}[W_1^{T^2}] - K \right) \end{aligned}$$

Since $\mathbb{E}[W_1^{T^2}] = 0$, the final price is:

$$\text{Price} = P(0; 2) \left[\ln(F(0)) - K - \frac{1}{2}\sigma^2 \right]$$

- (b) The price does depend on σ . Unlike a standard linear forward contract (where volatility washes out), this payoff is concave. The price is decreasing in σ . As volatility increases, the "concavity drag" ($-\frac{1}{2}\sigma^2$) becomes more negative, lowering the contract value. You would obtain σ from the At-The-Money (ATM) Caplet volatility surface, specifically looking at the 1-year into 1-year volatility (the 1×2 caplet).

3. Single-Curve Interest Rate Swap

Suppose that we are in a single curve framework, the term structure of interest rates is defined by the following function:

$$L(0, t) = 1.3\% + 1.1\%(t)$$

you enter a 3y receiver vanilla swap 1y forward, fixed and floating payments are annual, and the notional is 100 mln.

- what is the fixed rate of the swap?
- what is the relation between FRAs and Swaps? In this case, can you say how single forwards NPV are distributed across the life of the swap?
- Is the position short or long interest rates? What is the difference with the multi curve framework?
- Suppose that, as soon as we execute the trade, all the swap rate curve has a parallel shift of -1% . Compute the value of the derivative after the shift.

Solution:

We are given the term structure:

$$L(0, t) = 0.013 + 0.011t$$

and a 3-year receiver swap starting in 1 year, with annual payments and notional

$$N = 100,000,000.$$

The payment dates are:

$$T_0 = 1, \quad T_1 = 2, \quad T_2 = 3, \quad T_3 = 4.$$

Assuming a single-curve framework with simple compounding:

$$P(0, t) = \frac{1}{1 + tL(0, t)}.$$

Step 1: Discount factors

$$P(0, 1) = 0.9765625, \quad P(0, 2) = 0.934579, \quad P(0, 3) = 0.878735, \quad P(0, 4) = 0.814332.$$

(a) Fixed rate of the swap

The swap annuity is:

$$A(0) = P(0, 2) + P(0, 3) + P(0, 4) = 2.627646.$$

The forward swap rate is:

$$K = \frac{P(0, 1) - P(0, 4)}{A(0)} = \frac{0.9765625 - 0.814332}{2.627646} \approx 0.061740.$$

$$\boxed{K \approx 6.174\%}$$

(b) Relation between FRAs and swaps

A vanilla swap can be replicated as a portfolio of forward rate agreements (FRAs) on the periods:

$$[1, 2], \quad [2, 3], \quad [3, 4].$$

The corresponding forward rates are:

$$F(0; 1, 2) \approx 4.492\%, \quad F(0; 2, 3) \approx 6.355\%, \quad F(0; 3, 4) \approx 7.909\%.$$

Since the curve is upward sloping:

$$F(0; 1, 2) < K < F(0; 2, 3) < F(0; 3, 4).$$

Therefore, for a receiver swap:

- the first FRA has positive NPV ($K > F(0; 1, 2)$),
- the later FRAs have negative NPV ($K < F(0; 2, 3)$, $K < F(0; 3, 4)$).

The individual NPVs sum to zero at inception.

(c) Interest rate exposure and multi-curve framework

A receiver swap (receive fixed, pay floating) benefits from a decrease in interest rates, since the fixed leg becomes relatively more valuable. Therefore:

$$\boxed{\text{Receiver swap} = \text{short interest rates (long duration)}}$$

In a single-curve framework:

- the same curve is used for discounting and forward rates,
- the floating leg simplifies to

$$PV_{\text{float}} = N(P(0, T_0) - P(0, T_3)).$$

In a multi-curve framework:

- discounting and forwarding use different curves,
- the floating leg must be computed as the sum of discounted forward coupons,
- it no longer simplifies to a difference of discount factors.

(d) Value after a parallel shift of -1%

Since the exercise specifies a parallel shift of the swap rate curve, we can use the swap annuity as sensitivity:

$$V_{\text{receiver}} \approx N \cdot A(0) \cdot (K - S_{\text{new}})$$

A -1% shift implies:

$$S_{\text{new}} = K - 1\% \quad \Rightarrow \quad K - S_{\text{new}} = 1\% = 0.01.$$

Therefore:

$$V_{\text{receiver}} \approx 100,000,000 \times 2.627646 \times 0.01$$

$$V_{\text{receiver}} \approx 2,627,646.$$

$$V \approx +2.628 \text{ million}$$

The value is positive because the position is a receiver swap and interest rates have decreased.

4. Solve the Hull–White Process

Consider the one-factor Hull–White model (extended Vasicek) under the risk neutral measure $\mathbb{Q}$:

$$dr(t) = \kappa(\theta(t) - r(t))dt + \sigma dW_t, \quad r(0) = r_0,$$

where $\kappa > 0$ and $\sigma > 0$ are constants, while $\theta(t)$ is a deterministic function chosen to fit the initial term structure

$$\theta(t) = f^M(0, t) + \frac{1}{\kappa} \frac{\partial}{\partial t} f^M(0, t) + \frac{\sigma^2}{2\kappa^2} (1 - e^{-2\kappa t}),$$

where $f^M(0, t)$ denotes the market instantaneous forward rate at time 0 for maturity t .

- Solve the stochastic differential equation for $r(t)$ and derive the conditional expectation $\mathbb{E}^{\mathbb{Q}}[r(t) \mid \mathcal{F}_s]$ and conditional variance $\text{Var}^{\mathbb{Q}}[r(t) \mid \mathcal{F}_s]$, for $0 \leq s < t$.
- What is the difference between the Hull–White model and the Vasicek model?

- (c) Briefly describe Credit Valuation Adjustment (CVA). Explain how you would value the CVA of an interest rate swap using the Hull–White model. Why is an interest rate model needed even though the swap payoff is linear in interest rates?

Solution:

a) Define

$$X(t) = f(t, r(t)) := e^{\kappa t} r(t).$$

We apply Itô's lemma to the function $f(t, r) = e^{\kappa t} r$. Its derivatives are

$$\frac{\partial f}{\partial t}(t, r) = \kappa e^{\kappa t} r, \quad \frac{\partial f}{\partial r}(t, r) = e^{\kappa t}, \quad \frac{\partial^2 f}{\partial r^2}(t, r) = 0.$$

Since

$$dr(t) = \kappa(\theta(t) - r(t))dt + \sigma dW_t,$$

Itô's lemma gives

$$dX(t) = \frac{\partial f}{\partial t}(t, r(t))dt + \frac{\partial f}{\partial r}(t, r(t))dr(t) + \frac{1}{2} \frac{\partial^2 f}{\partial r^2}(t, r(t))\sigma^2 dt.$$

Because $\frac{\partial^2 f}{\partial r^2} = 0$, this becomes

$$dX(t) = \kappa e^{\kappa t} r(t) dt + e^{\kappa t} dr(t).$$

Substituting the dynamics of $r(t)$,

$$\begin{aligned} dX(t) &= \kappa e^{\kappa t} r(t) dt + e^{\kappa t} [\kappa(\theta(t) - r(t))dt + \sigma dW_t] \\ &= \kappa e^{\kappa t} \theta(t) dt + \sigma e^{\kappa t} dW_t. \end{aligned}$$

Hence,

$$d(e^{\kappa t} r(t)) = \kappa e^{\kappa t} \theta(t) dt + \sigma e^{\kappa t} dW_t.$$

Integrating from s to t ,

$$e^{\kappa t} r(t) - e^{\kappa s} r(s) = \int_s^t \kappa e^{\kappa u} \theta(u) du + \sigma \int_s^t e^{\kappa u} dW_u.$$

Therefore,

$$r(t) = e^{-\kappa(t-s)} r(s) + \kappa \int_s^t e^{-\kappa(t-u)} \theta(u) du + \sigma \int_s^t e^{-\kappa(t-u)} dW_u.$$

Taking conditional expectation with respect to $\mathcal{F}_s$, and using that the stochastic integral has zero conditional mean,

$$\mathbb{E}^{\mathbb{Q}}[r(t) \mid \mathcal{F}_s] = e^{-\kappa(t-s)} r(s) + \kappa \int_s^t e^{-\kappa(t-u)} \theta(u) du.$$

For the conditional variance, only the stochastic integral contributes. By Itô isometry,

$$\begin{aligned} \text{Var}^{\mathbb{Q}}[r(t) \mid \mathcal{F}_s] &= \sigma^2 \int_s^t e^{-2\kappa(t-u)} du \\ &= \frac{\sigma^2}{2\kappa} (1 - e^{-2\kappa(t-s)}). \end{aligned}$$

Thus, conditionally on $\mathcal{F}_s$, the short rate is Gaussian with mean

$$e^{-\kappa(t-s)}r(s) + \kappa \int_s^t e^{-\kappa(t-u)}\theta(u) du$$

and variance

$$\frac{\sigma^2}{2\kappa} (1 - e^{-2\kappa(t-s)}) .$$

b) The Hull–White model is an extended version of the Vasicek model.

In the Vasicek model, the short rate follows

$$dr(t) = \kappa(\theta - r(t))dt + \sigma dW_t,$$

where θ is constant. This implies that the model is not flexible enough to fit exactly a generic observed initial term structure.

In the Hull–White model, instead, the dynamics are

$$dr(t) = \kappa(\theta(t) - r(t))dt + \sigma dW_t,$$

where $\theta(t)$ is deterministic and time-dependent. This time dependence allows the model to match the initial term structure exactly.

Therefore:

- Vasicek has constant long-run mean θ ;
- Hull–White replaces it with a time-dependent function $\theta(t)$;
- Hull–White fits the initial yield curve exactly, while Vasicek generally does not;
- both models are Gaussian and analytically tractable.

c) Credit Valuation Adjustment (CVA) is the adjustment to the risk-free value of a derivative that accounts for the possibility that the counterparty may default before the maturity of the contract. It represents the expected discounted loss due to counterparty credit risk.

In general, the unilateral CVA can be written as

$$\text{CVA} = (1 - R) \int_0^T \mathbb{E}^{\mathbb{Q}} [D(0, t) V^+(t)] d\text{PD}(0, t),$$

where R is the recovery rate, $D(0, t)$ is the stochastic discount factor from t to 0, $V^+(t) = \max(V(t), 0)$ is the positive exposure of the portfolio at time t , and $d\text{PD}(0, t)$ is the default probability over the infinitesimal interval around t .

In a discrete-time implementation, this is often approximated by

$$\text{CVA} \approx (1 - R) \sum_{i=1}^n \mathbb{E}^{\mathbb{Q}} [D(0, t_i) V^+(t_i)] \Delta\text{PD}(t_{i-1}, t_i),$$

where $\Delta\text{PD}(t_{i-1}, t_i)$ is the default probability between two consecutive dates.

To value the CVA of an interest rate swap using the Hull–White model, one would proceed as follows. First, the Hull–White model is calibrated to the initial yield

curve and, typically, to market prices of liquid interest rate options such as swaptions or caps/floors. Then, under the risk-neutral measure, one simulates future paths of the short rate $r(t)$. Along each simulated path and at each future exposure date t_i , the remaining swap is revalued using the simulated term structure implied by the Hull–White model. This gives a simulated mark-to-market value $V(t_i)$. The positive part,

$$V^+(t_i) = \max(V(t_i), 0),$$

is then averaged across paths to obtain the expected exposure or discounted expected exposure. Finally, the exposure profile is combined with the counterparty default probabilities and the loss given default $1 - R$.

The Hull–White model is needed even though the interest rate swap is linear in interest rates because CVA depends on the future positive exposure of the swap, not only on its current value. The exposure is nonlinear due to the positive-part operator:

$$V^+(t) = \max(V(t), 0).$$

Therefore, even if the swap value is approximately linear in rates, the CVA depends on the distribution of future swap values. In particular, one needs a model for the stochastic evolution of the yield curve in order to compute future mark-to-market values, expected positive exposures, and their discounted expectation. The Hull–White model provides this distribution while remaining analytically tractable and calibrated to the initial term structure.